

Answer the following questions

1. In 16-bit 2's complement representation, the decimal number -28 is: [2 marks]

$$\underline{28} = \underline{2^4} + \underline{2^3} + 2^2 = 16 + 8 + 4$$

$$\underline{14} \ 0 \quad \underline{3} \ 1 \quad \Rightarrow 001111001111100100$$

2. What is the minimal form of the Karnaugh map shown below? Assume that X don't care terms. [2 marks]

$$F = \bar{C}\bar{D} + C\bar{B} + D\bar{B}$$

CD \ AB	01	11	10
00	1	X	1
01	1	1	1
11			
10	1		1

$$1) \bar{C}A\bar{B}$$

$$2) \bar{C}\bar{D}\bar{B}$$

$$3) \bar{C}A\bar{B}$$

$$F = \bar{C}A\bar{B} + \bar{C}\bar{D}\bar{B} + \bar{C}A\bar{B}$$

3. Convert the following Boolean expression into standard SOP [2 marks]

$$F(A, B, C, D) = ABC + \bar{B}\bar{A} + AB\bar{C}D$$

$$= A\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}CD + A\bar{B}CD$$

$$F = \bar{C}\bar{B} + C\bar{D}\bar{B} + A\bar{B}$$

4. Using 2's Complement Subtraction, find $(54)_{10} - (67)_{10}$ (show your calculations) [3 marks]

$$\left\{ \begin{array}{r} 54 = 110110 \\ 1 \ 101010 \\ \hline 1011110 \\ + 1 \\ \hline 1011111 \end{array} \right\} \quad \left\{ \begin{array}{r} 67 = 1000011 \\ 1 \ 100000 \\ \hline 1011111 \\ + 1 \\ \hline 10111000 \end{array} \right\} \quad \begin{array}{r} 110110 \\ - 10111000 \\ \hline 10111110 \end{array}$$

5. Consider the following function [2 marks]

$$F(P, Q, R, S) = \sum (0, 2, 5, 7, 8, 10, 13, 15). \text{ Simplify this function}$$

	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 ₀	1 ₂	0 ₃	1 ₈
$\bar{P}Q$	0 ₄	1 ₅	0 ₇	0 ₆
PQ	0 ₁₂	0 ₁₃	1 ₁₅	0 ₁₄
$P\bar{Q}$	1 ₁	0 ₉	0 ₁₁	1 ₁₀

$$F = Q\bar{S} + \bar{Q}\bar{S}$$

$$= Q \oplus \bar{S}$$

10. Consider the following function, $F = (A + BC)(\bar{A}\bar{B} + \bar{A}C)$, apply DE Morgan's theory. [2 marks]

$$= (A + BC) + (\bar{A}\bar{B} + \bar{A}C)$$

$$= \bar{A} \cdot (\bar{B} + \bar{C}) + (A + B) \cdot (\bar{A} + \bar{C})$$

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11. Using Boolean algebra, simplify the following function [3 marks]

$$F = [A\bar{B}(C + BD)\bar{B}\bar{A}]C$$

$$F = (\underbrace{A\bar{B}C + A\bar{B}BD}_{(0+0)}\bar{B}\bar{A})C$$

$$[(0+0)\bar{B}A]C$$

$$[0]C$$

$$= 0$$

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12. Simplify and sketch the following function. [3 marks]

$$F(A, B, C, D) = \sum(1, 2, 5, 15), \quad F(A, B, C, D) = \sum(7, 8, 10, 14)$$

	$\bar{C}D$	$\bar{C}\bar{D}$	$C\bar{D}$	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	11	10

$$F = BD + \bar{B}C\bar{D} + \bar{A}\bar{C}D$$

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13. Consider the binary number $(100101)_2$, Find the gray code equivalent to this number. [2 marks]

$$G \Rightarrow 110111$$

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